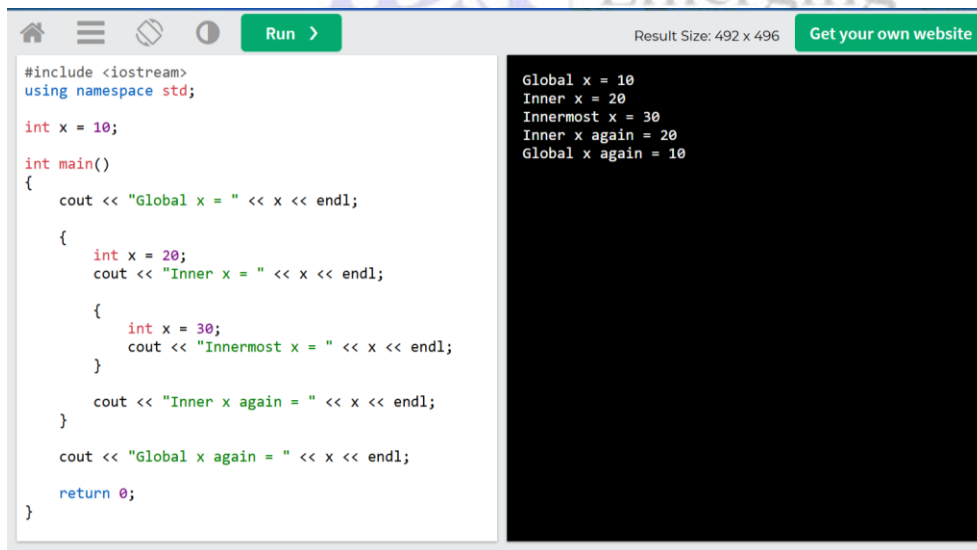


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Language	Imperative	Declarative	Von Neumann	OOP	Functional	3GL	4GL	Scripting
C	✓		✓			✓		
Lisp					✓	✓		
C++	✓		✓	✓		✓		
ML					✓	✓		
COBOL	✓		✓			✓		
FORTRAN	✓		✓			✓		
Perl	✓					✓		✓
Python	✓			✓	✓	✓		✓
Java	✓		✓	✓		✓		
VB	✓		✓	✓		✓		

Lab 1 — Solved Tasks

Task A — Language Classification



```

#include <iostream>
using namespace std;

int x = 10;

int main()
{
    cout << "Global x = " << x << endl;

    {
        int x = 20;
        cout << "Inner x = " << x << endl;

        {
            int x = 30;
            cout << "Innermost x = " << x << endl;
        }

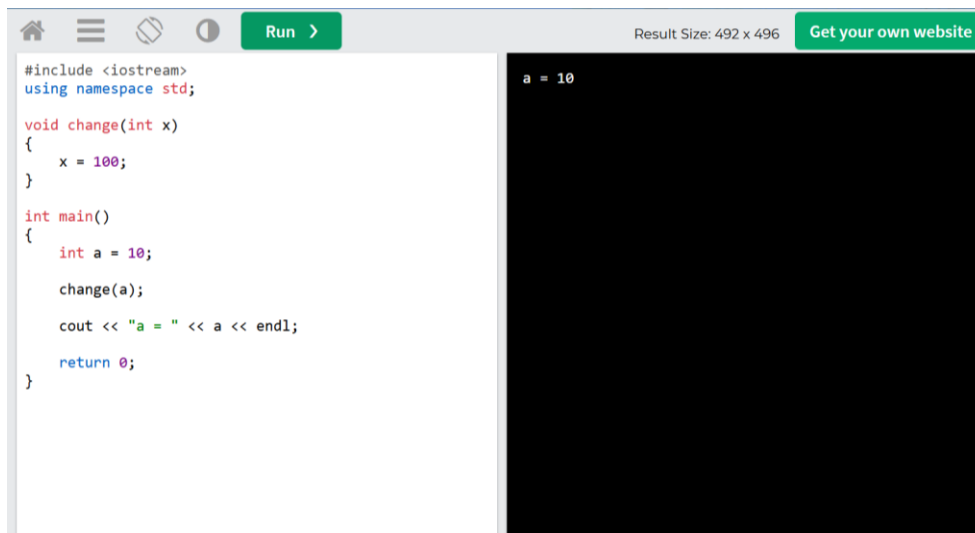
        cout << "Inner x again = " << x << endl;
    }

    cout << "Global x again = " << x << endl;

    return 0;
}
  
```

Global x = 10
 Inner x = 20
 Innermost x = 30
 Inner x again = 20
 Global x again = 10

Task B — Scope and Shadowing



```
#include <iostream>
using namespace std;

void change(int x)
{
    x = 100;
}

int main()
{
    int a = 10;

    change(a);

    cout << "a = " << a << endl;

    return 0;
}
```

Result Size: 492 x 496

Get your own website

a = 10

Task C — Pass by Value



```
#include <iostream>
using namespace std;

void change(int &x)
{
    x = 100;
}

int main()
{
    int a = 10;

    change(a);

    cout << "a = " << a << endl;

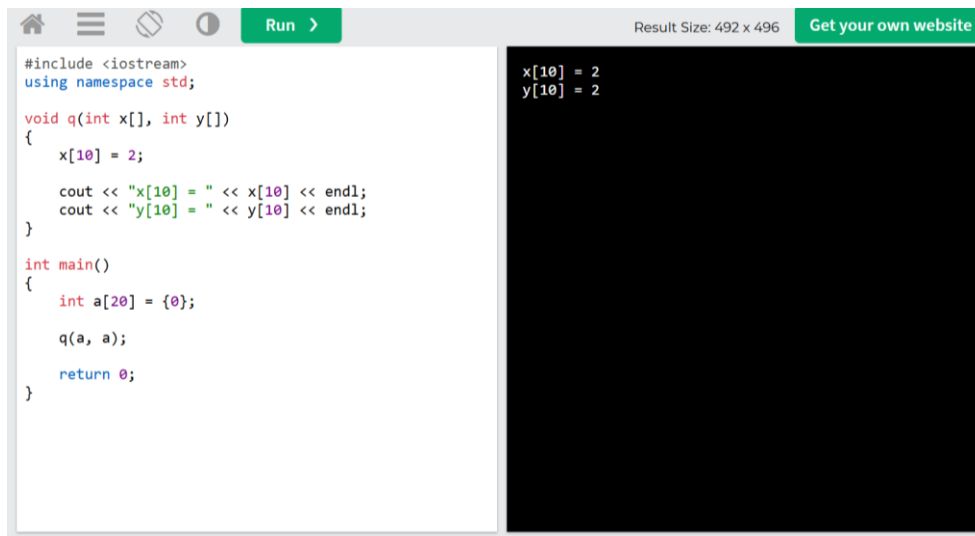
    return 0;
}
```

Result Size: 492 x 496

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a = 100

Task D — Pass by Reference



```
#include <iostream>
using namespace std;

void q(int x[], int y[])
{
    x[10] = 2;

    cout << "x[10] = " << x[10] << endl;
    cout << "y[10] = " << y[10] << endl;
}

int main()
{
    int a[20] = {0};

    q(a, a);

    return 0;
}
```

Result Size: 492 x 496

Get your own website

x[10] = 2
y[10] = 2

Task E — Pass by Name

procedure P(x)

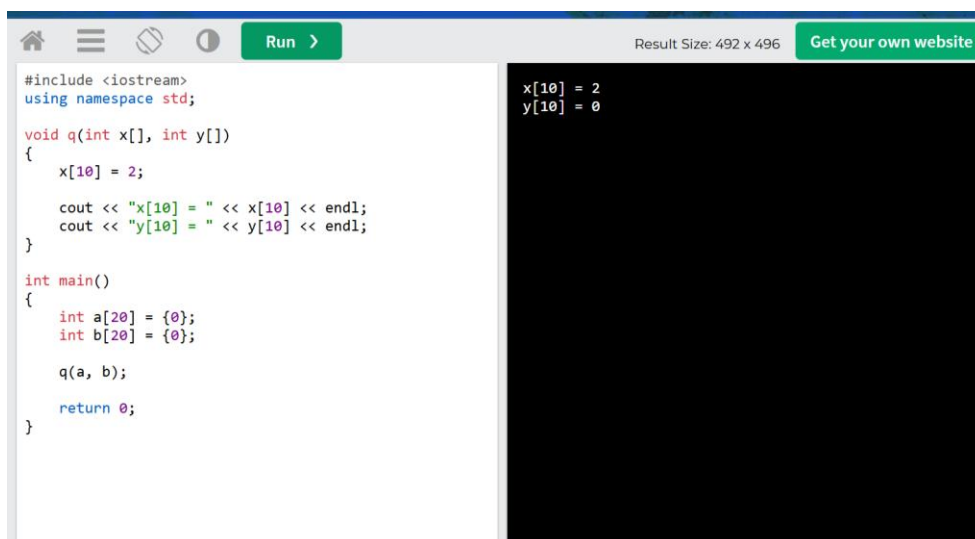
```
{
    print(x);
}
```



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P(a + b);

Task F — Aliasing



```
#include <iostream>
using namespace std;

void q(int x[], int y[])
{
    x[10] = 2;

    cout << "x[10] = " << x[10] << endl;
    cout << "y[10] = " << y[10] << endl;
}

int main()
{
    int a[20] = {0};
    int b[20] = {0};

    q(a, b);

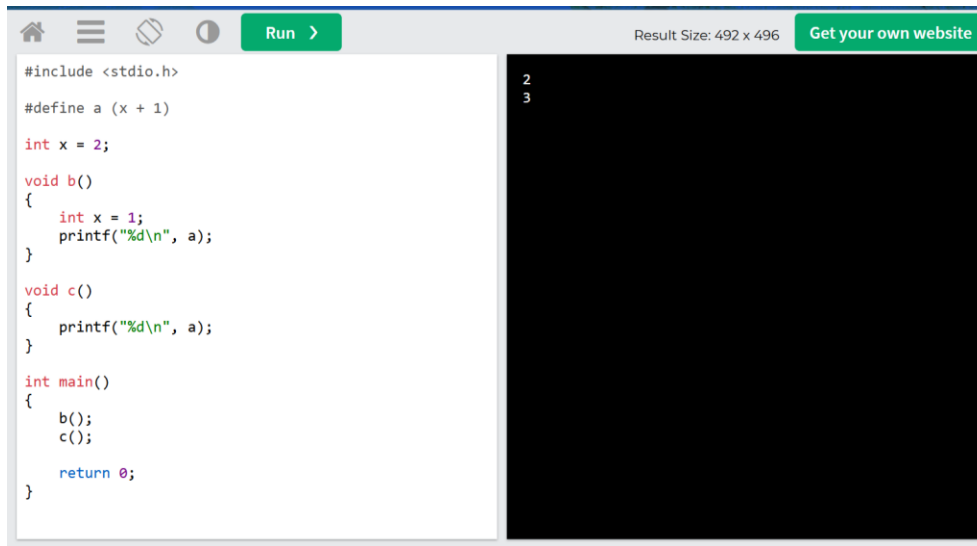
    return 0;
}
```

Result Size: 492 x 496

Get your own website

x[10] = 2
y[10] = 0

Task G — Macro Expansion



The screenshot shows an online C compiler interface. On the left, the source code is displayed with syntax highlighting. It includes `<stdio.h>`, defines a macro `a (x + 1)`, declares `int x = 2;`, and defines two functions: `b()` which sets `x = 1` and prints `a`, and `c()` which prints `a`. The `main()` function calls `b()` and `c()` before returning 0. On the right, the output window shows the results of the program execution, which are the numbers 2 and 3 on separate lines. The top of the interface includes navigation icons, a 'Run' button, and a 'Get your own website' button.

```
#include <stdio.h>

#define a (x + 1)

int x = 2;

void b()
{
    int x = 1;
    printf("%d\n", a);
}

void c()
{
    printf("%d\n", a);
}

int main()
{
    b();
    c();

    return 0;
}
```

2
3

Task H — Compare macro.c and macro.i

macro.c



Preprocessor



macro.i

